

Marketing Funnel & Conversion Performance Analysis Report

Executive Summary

This report presents a comprehensive analysis of marketing funnel performance using Python-based Exploratory Data Analysis (EDA) and Power BI dashboarding techniques. The objective of the project is to evaluate customer progression through different marketing funnel stages, identify conversion bottlenecks, measure campaign effectiveness, and optimize marketing ROI.

The project combines data analytics, KPI measurement, funnel analysis, and interactive visualizations to transform raw marketing data into meaningful business insights.

Project Objectives

- Analyze overall marketing funnel performance
 - Identify customer drop-off points
 - Measure marketing channel effectiveness
 - Evaluate campaign performance
 - Calculate business KPIs
 - Build an interactive Power BI dashboard
 - Generate actionable business recommendations
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Dataset Description

The dataset contains campaign-level marketing performance data with the following fields:

Column	Description
Date	Campaign date
Channel	Marketing platform
Campaign	Campaign name
Impressions	Number of users reached
Clicks	Number of clicks
Leads	Number of leads generated

Column	Description
Conversions	Successful conversions
Revenue	Revenue generated
Cost	Marketing expenditure
Device	Customer device type
Region	Geographic region

Tools & Technologies Used

Python Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn

Visualization Tool

- Power BI

Development Environment

- Jupyter Notebook / Google Colab
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Data Cleaning & Preparation

The dataset was cleaned and transformed before analysis.

Steps Performed

Missing Value Handling

- Null values were identified and replaced.

Duplicate Removal

- Duplicate records were removed.

Data Type Conversion

- Date columns converted into datetime format.

Feature Engineering

Additional KPI columns were created:

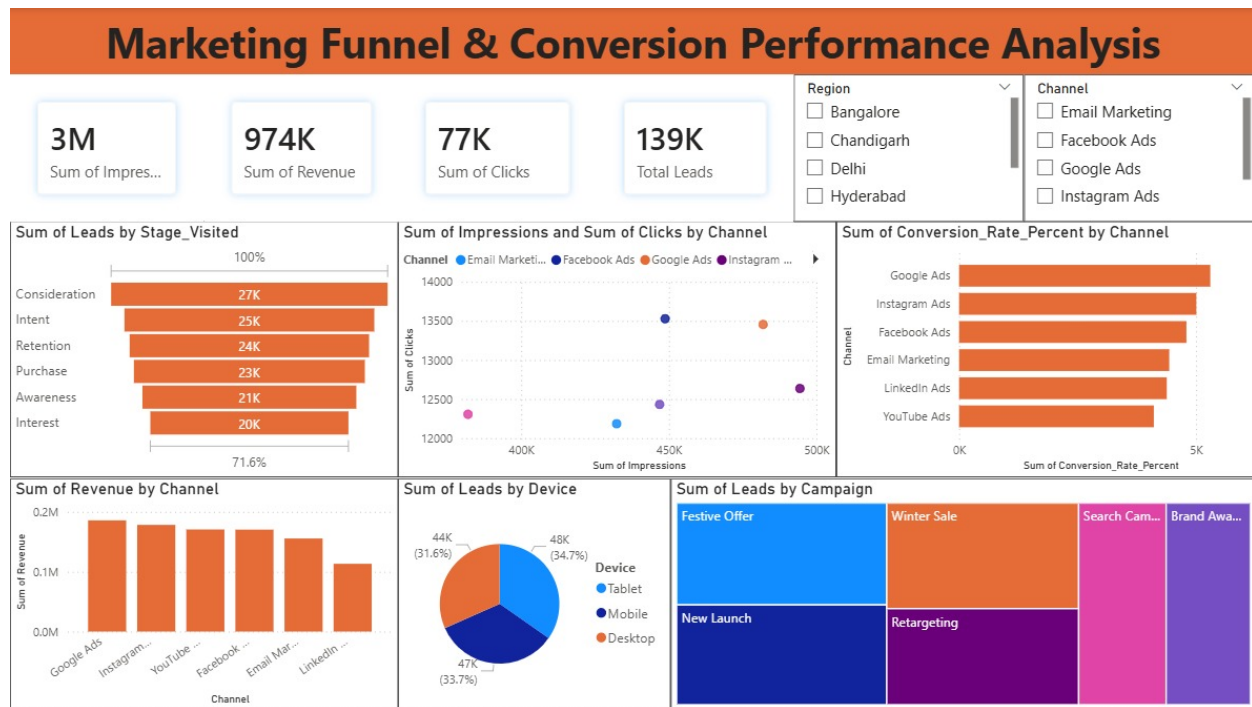
- CTR

- Conversion Rate
- CPL
- CAC
- ROI

Key Performance Indicators (KPIs)

KPI	Formula
CTR	Clicks / Impressions
Conversion Rate	Conversions / Leads
CPL	Cost / Leads
CAC	Cost / Conversions
ROI	(Revenue – Cost) / Cost

Power BI Dashboard



Dashboard Snapshot

The following Power BI dashboard was developed for interactive business intelligence reporting.

Dashboard Features

KPI Cards

- Total Impressions: 3M
- Total Revenue: 974K
- Total Clicks: 77K
- Total Leads: 139K

Funnel Analysis

The funnel chart visualizes customer progression across:

- Awareness
- Interest
- Consideration
- Intent
- Purchase
- Retention

Channel Performance Analysis

Comparison of:

- Revenue by Channel
- Conversion Rate by Channel
- Impressions and Clicks by Channel

Campaign Performance

Treemap visualization comparing:

- Festive Offer
- Winter Sale
- New Launch
- Retargeting
- Search Campaign
- Brand Awareness

Device Analysis

Lead generation analyzed across:

- Mobile
- Desktop
- Tablet

Interactive Filters

- Region Filter
 - Channel Filter
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Dashboard Insights

Major Insights

Conversion Bottleneck

A significant drop-off occurs between:

Leads → Conversions

High Performing Channels

Google Ads and Instagram Ads generated the highest revenue and conversion rates.

Campaign Performance

Festive Offer and Winter Sale campaigns contributed the majority of leads.

Device Insights

Mobile and tablet devices generated the highest customer engagement.

Revenue Contribution

Google Ads generated the highest overall revenue.

Exploratory Data Analysis (EDA)

Univariate Analysis

- Distribution analysis for impressions, clicks, leads, and conversions.
- Outlier detection using boxplots.

Bivariate Analysis

- Revenue vs Cost analysis.
- Clicks vs Conversion relationship.

Correlation Analysis

Strong relationships identified between:

- Cost and Revenue
 - Leads and Conversions
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Business Recommendations

1. Optimize Conversion Funnel

- Improve landing pages
- Simplify customer journey
- Enhance call-to-action strategy

2. Increase Budget for High ROI Channels

- Focus investment on Google Ads and Instagram campaigns.

3. Improve Audience Targeting

- Use segmentation strategies
- Retarget high-intent customers

4. Reduce Low Performing Campaigns

- Pause campaigns with poor ROI.

5. Continuous Dashboard Monitoring

- Track KPIs regularly using Power BI.
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Conclusion

This project demonstrates how Python analytics and Power BI dashboards can transform marketing data into actionable business intelligence.

The analysis successfully identified:

- Funnel bottlenecks
- High-performing channels
- Revenue-driving campaigns
- Customer engagement patterns

The dashboard enables organizations to make data-driven decisions, optimize marketing spend, improve conversion performance, and maximize business revenue.

Future Enhancements

- Machine Learning conversion prediction
- Customer segmentation analysis
- Real-time dashboard integration
- Forecasting revenue trends

- Advanced customer behavior analytics
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Final Deliverables

- ✓ Cleaned Dataset
 - ✓ Python EDA Notebook
 - ✓ KPI Metrics
 - ✓ Power BI Dashboard
 - ✓ Business Insights Report
 - ✓ README Documentation
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Author

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